Performance Testing

Date	29 september 2026
Team ID	6ab2abc01266c775575a9dda
Project Name	Farm Management System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	API Performance Testing / Response Time Testing Load Testing and Stress Testing
Target Module / API	User Registration API User Login API Farm Management API Crop Management API
Test Environment	Localhost / Local Development Environment
Test Date	**29 September 2026**

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login API performance test	10	60	Login requests should receive responses within acceptable time. Registration requests should be processed successfully without errors.
2	User Registration API performance test	10	60	

3	Farm Details API performance test	10	60	Farm data should be retrieved/updated with acceptable response time.
4	Crop Management API performance test	10	60	Crop information should be processed successfully under load.



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Average response time is within target.
2	Response Time (Max)	< 5 seconds	2.8 sec	Pass	Maximum response time is within acceptable limit.
3	Throughput (Req/sec)		8.5 req/sec	Pass	Requests were processed consistently.
4	Error Rate	< 1%	0.2%	Pass	Error rate is below the target.

5 CPU Utilization < 80%

6 Memory Utilization < 80%

Step 4: Observations & Analysis

The Farm Management System performed within the expected response time during testing. The major APIs responded within the acceptable performance limits. The system was able to process multiple requests without significant delays.

No major performance bottlenecks were identified during the testing period.

API requests and database operations were reviewed to avoid unnecessary processing. Input validation and efficient data retrieval were used to improve the overall response time.

Step 5: Screenshots / Evidence
Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Postman API response showing response time and successful response.

JMeter performance test report showing response time, throughput and error rate.